

spoilsport to the advent of FOSS.

The easy solution to this problem is to put up a server inside the institute or organisation, where all the contents are downloaded and updated periodically, and users get the software from this local server instead of the Internet. Considering that the cost of bandwidth inside a LAN is trivial and usually offers much better throughput, mirroring can be an ideal solution to reduce expenditure and can considerably speed up installations of new software or updates. It can even reduce the need for physical media like a CD or DVD, as you can use the server for disk-less network installations.

In the subsequent sections, I will take you through the steps of setting up a Fedora mirror.

Mirroring requirements

Hardware

Mirroring does not cost much as far as hardware is concerned. If you are going to mirror the whole Fedora content, at least 1TB of disk space is needed. But if you are not an ISP or a big educational institute, you probably won't need all the content available in the Fedora repositories. It should be fine for most organisations to keep 32-bit and 64-bit repositories of the last two releases, along with their updates. For example, if you are mirroring right now, it would be good to keep 32-bit (generally called x86) and 64-bit (x86_64) repositories of Fedora 10 and 11, along with their updates.

A server with approximately 250 GB of hard disk space (though the actual need will depend upon the content you want to keep), and 2 – 4 GB of RAM should do perfectly.

Software

The software requirements for mirroring are also minimal. All you need is an Apache Web server or an FTP server. However, please check your version of Apache. If the version is 1.x or 2.0, you will need both the Apache and the FTP server, because earlier Apache servers cannot handle files over 2 GB in size; so you have to redirect the ISO download requests to the FTP server. However, if you are using Apache 2.1 or 2.2, you need not worry about this as large file handling support has been added in these versions. Here, we will explore mirroring only using Apache. Mirroring with FTP is similar and needs no remarkably different configuration.

Bandwidth

The most essential requirement for mirroring is bandwidth. How long your download will take depends on the available bandwidth. Mirroring over a 5 MBps leased line may take several days for each release being mirrored, but most of these contents need to be downloaded only once. The subsequent downloads will need much less bandwidth, often as minimal as a couple of hundred megabytes per day.

If you are trying to be listed as a public mirror of Fedora, by which you want to offer downloads to people outside your organisation, the official bandwidth requirement is 100 MBps. However, in countries like

India, where very few public mirrors are available, this requirement is often relaxed. The first public Fedora mirror in India started with a 5MBps leased line, until other institutes like NIT-H, IIT-M and IIT-K stepped in.

What to mirror?

Though I have suggested that you choose to mirror the last two releases along with their updates, this would obviously depend on you. The complete list of directories, along with their sizes, is given at http://download.fedora.redhat.com/pub/DIRECTORY_SIZES.txt. You can choose what to mirror and what not to, depending on your organisational needs.

Public or private

It's also very important to decide if you want to make your mirror a public mirror, which serves content to people outside your organisation, or a private mirror that serves people only inside your organisation. If you don't have large bandwidth, at least ~100Mbps, it is better to go for a private mirror. However, for countries like India, where the number of mirrors are far less than required, you can go public with 15-20 MBps bandwidth.

The mirroring procedure

Having finished with the requirements, we now move on to actually setting up a Fedora mirror. Before you get your hands dirty, it would be better if you could study the directory structure of the Fedora repository for a while. You can find that at <http://download.fedora.redhat.com/pub/fedora/linux/>

Synchronising content

Synchronising content is, to put it simply, copying the content of a Fedora mirror into your server in such a way that all the properties of the files and directories being transferred remain unchanged. As this is the most time-consuming process involving a large number of file downloads, it is suggested that you first get this started and while it pulls content from the server, you do other necessary configurations. The only reliable way to do mirroring is to use `rsync`, which is a utility for incremental file transfer. Like FTP, `rsync` also transfers files between a server and a client, but if the file transfer breaks down midway as a result of a network or power outage, it will resume transferring files from the point where it left off. From now on, we shall use the terms *synchronise* or *pull* instead of 'file transfer'.

It is best to set up a new user account on your system, which will perform the synchronisation.

```
# useradd -r -m mirror
```

The directory structure you are mirroring should match that of Fedora's master mirrors. To do so, simply create them and give your *mirror* user write permissions:

```
# mkdir -p /var/www/html/pub/fedora/linux/releases
```